Mini Project Report

on

Premium SaaS Invoice Management System

Submitted by

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Submitted To

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1.Objective

The primary objective of this project is to develop a secure, full-stack Software-as-a-Service (SaaS) application that simplifies the billing process for freelancers and businesses. The system aims to provide a seamless interface for users to securely register, dynamically generate professional PDF invoices, manage client billing history, and visualize revenue analytics in real-time.

2.Introduction

In today's fast-paced business environment, manual invoicing is time-consuming and prone to errors. This project addresses this issue by providing a comprehensive Invoice Management System built on the modern MERN architecture (MongoDB, Express.js, React.js, Node.js).

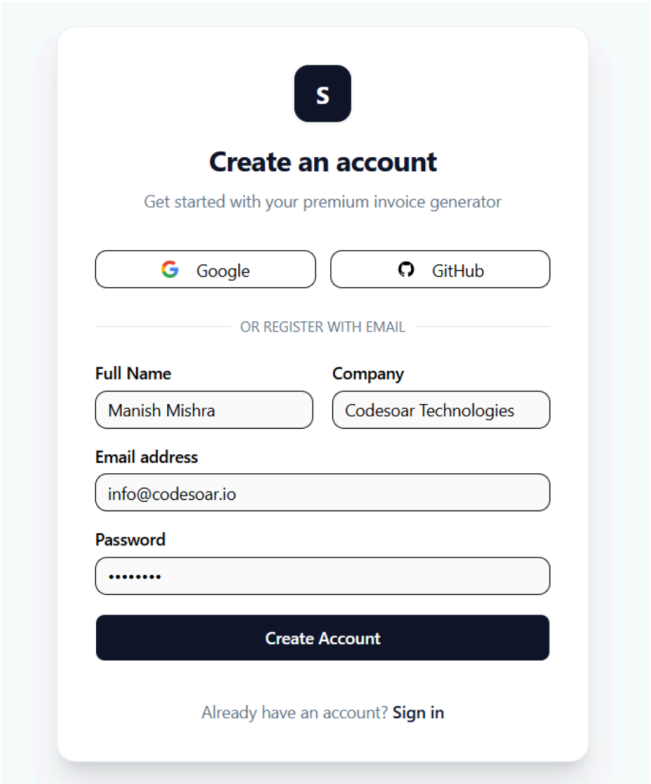
The application is designed to be a fully functional RESTful API and CRUD application. It features a stateless authentication system using JSON Web Tokens (JWT) to ensure user data privacy. Users can customize their business profiles, select localized currencies (e.g., USD, INR, EUR), add line items with automatic tax and discount calculations, and generate pixel-perfect PDF documents directly within the browser. Furthermore, an integrated dashboard provides real-time financial tracking, calculating pending and paid amounts automatically.

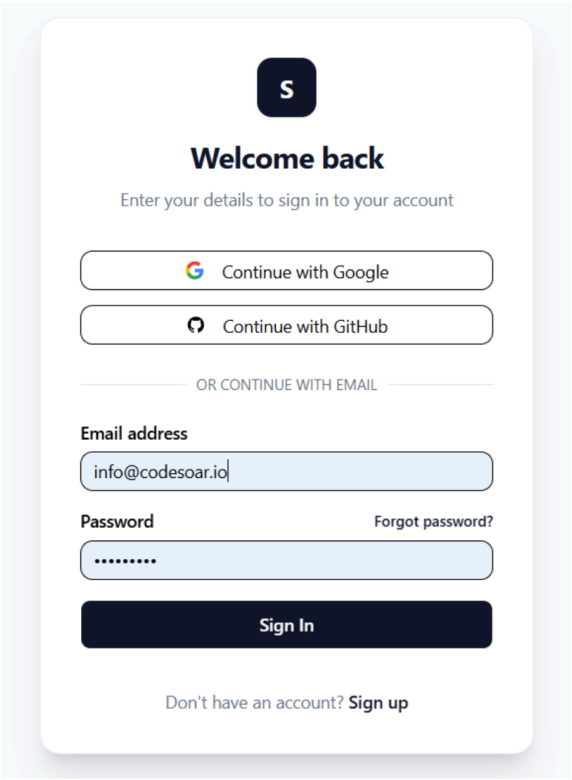
3.Tools and Technologies Used

Category	Technology	Purpose / Description
Frontend Technologies	React.js (via Vite)	For building a fast, component-based user interface.
	Tailwind CSS	For rapid, responsive UI styling.
	Shadcn UI & Radix UI	For accessible, premium interactive components.
	Zustand	For lightweight global state management.
Backend Technologies	Node.js & Express.js	To build the RESTful API and handle server-side routing.
	MongoDB & Mongoose	As the NoSQL database and Object Data Modeler (ODM) to store user profiles and invoice data.
Key Libraries & Tools	Bcrypt.js & JsonWebToken (JWT)	For secure password hashing and user authentication.

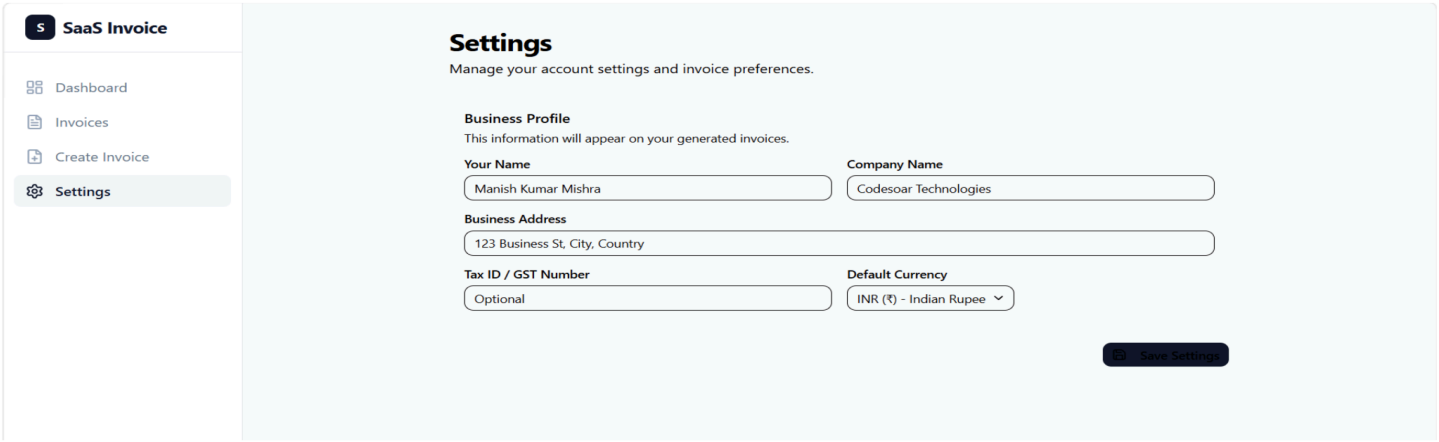
	@react-pdf/renderer	For client-side generation and downloading of PDF invoices.
	Recharts	For rendering dynamic financial charts on the dashboard.

4. Input Screenshots

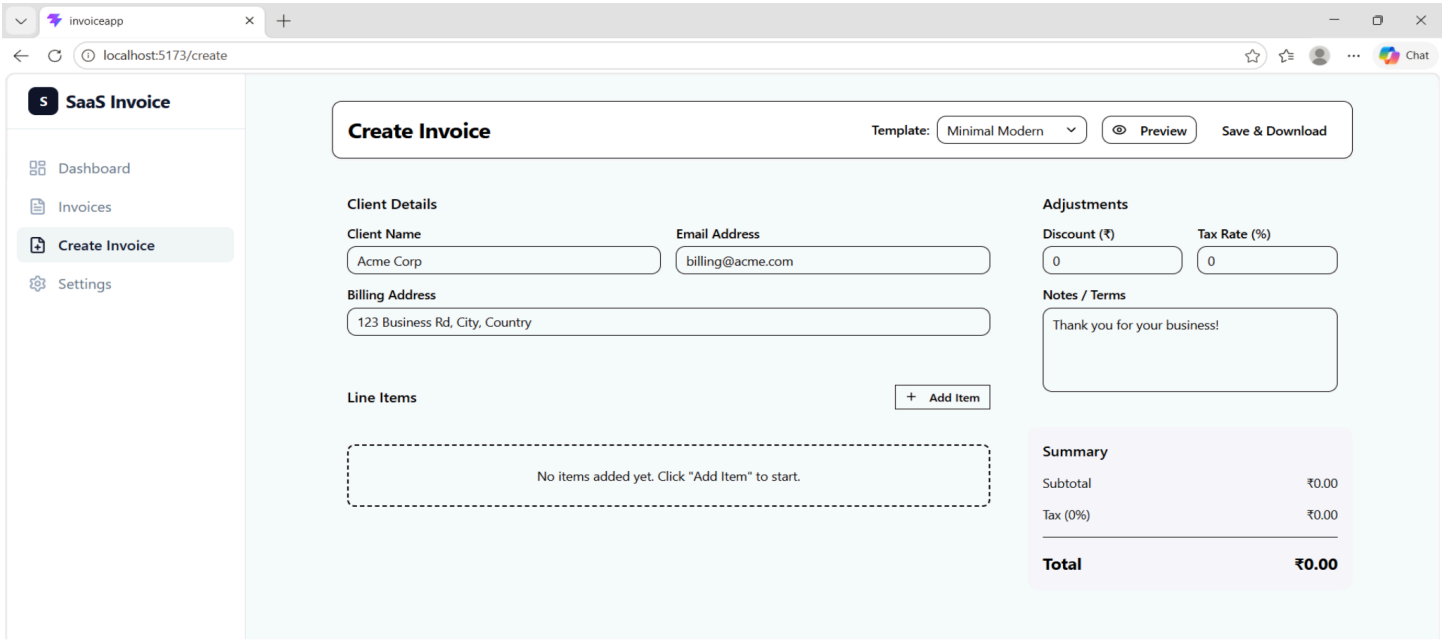




[Figure 1]
User Authentication Interface. This form captures the user's initial registration details (Name, Email, Password, Company) securely hashing the data before saving it to the database



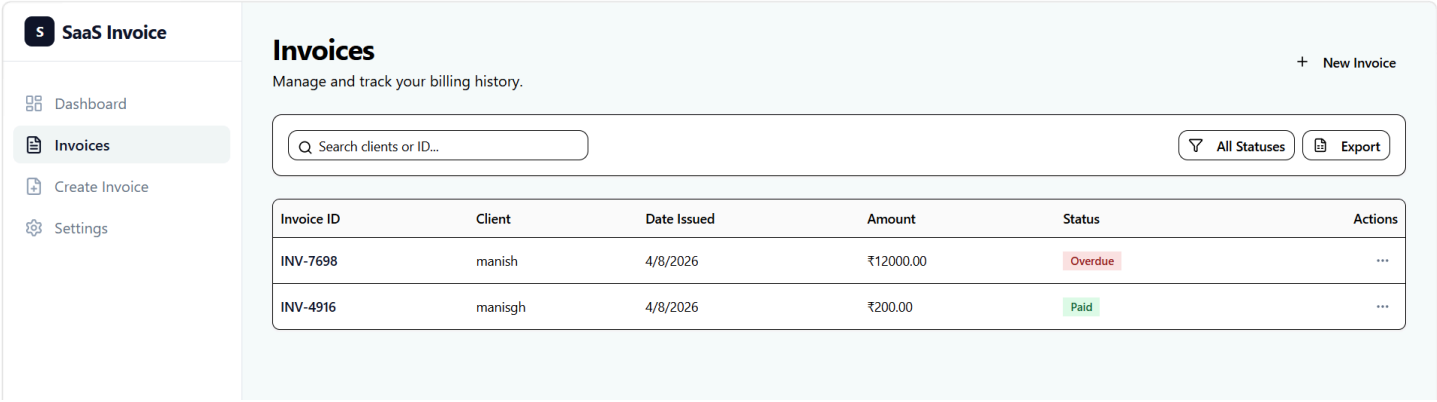
[Figure 2]
User Profile and Settings. This interface allows users to input their dynamic business details, including GST numbers and preferred default currency (e.g., INR, USD).



[Figure 3]
Invoice Generation Form. This input screen collects client details, dynamic line items (Quantity, Price), tax rates, and discount adjustments to calculate the final billing amounts.

5. Output Screenshots

[Figure 4]
Real-time Analytics Dashboard. The output view displaying calculated total revenue, pending amounts, paid invoice counts, and an interactive area chart based on the user's database records.



[Figure 5]
Invoice Management Data Table. The output interface where users can read, filter, update statuses (Paid/Pending), and delete saved invoices.

INVOICE

Invoice #: INV-4322

Date: 4/12/2026

Codesoar
Manish Kumar Mishra

Billed To:
Mithila Web

admin@mithilaweb.com

847239,Madhubani,Bihar

Description	Qty	Price	Amount
Web Design Services	1	₹15000.00	₹15000.00
Domain	1	5000.00	5000.00
Hosting	1	₹7500.00	₹7500.00

Subtotal:	₹27500.00
Discount:	-10.00
Tax (18%):	₹4948.20
Total Due:	**₹32438.20**

Notes & Terms:
Thank you for your business!

Thank you for your business. Generated by SaaS Invoice.

[Figure 6]
Generated PDF Document. The final downloadable output rendered via `@react-pdf/renderer`, showcasing the user's customized data, calculated totals, and selected currency symbol.

6.References

Github:https://github.com/official-Manish82/invoice_app

- **React Documentation: *react.dev* - For component lifecycle and hooks implementation.**
- **Express & Node.js Documentation: *expressjs.com* - For REST API routing guidelines.**
- **MongoDB Atlas & Mongoose: *mongoosejs.com* - For schema design and database integration.**
- **React-PDF Renderer: *react-pdf.org* - For generating complex PDF documents in standard React syntax.**
- **Tailwind CSS: *tailwindcss.com* - For UI component styling.**

7. Learning Outcome

Through the successful development of this project, several key technical and architectural concepts were mastered:

- **Full-Stack Integration:** Successfully bridged a React frontend with a Node.js/Express backend using CORS and RESTful API endpoints.
- **Database Management:** Learned to design NoSQL schemas and perform complex CRUD (Create, Read, Update, Delete) operations using MongoDB.
- **Web Security:** Gained practical experience in securing applications by hashing passwords with Bcrypt and protecting private routes using JWT-based authorization.
- **Advanced React Concepts:** Learned to manage complex local states, utilize useEffect for data fetching, and manage global states without prop-drilling using Zustand.
- **File Handling:** Understood how to map JSON data dynamically to generate and trigger the download of PDF files directly within the client browser.